

FOODFORWARD

A Web-Based Food Donation Platform
Connecting Donors, Restaurants, and Charities

PROJECT UPDATE REPORT

Course: Junior Project

Project Type: Web Application

Submitted To:

[Supervisor/Teacher Name]

[Department Name]

[University Name]

Submitted By:

SL	Team Member Name	Student ID	Role
1	[Student Name 1]	[Student ID]	Project Coordinator / Backend Developer
2	[Student Name 2]	[Student ID]	Frontend Developer / UI Designer
3	[Student Name 3]	[Student ID]	Database / Testing / Documentation

Submission Date: [Date]

TABLE OF CONTENTS

1. Executive Summary
2. Introduction
 - 2.1 Project Background
 - 2.2 Problem Statement
 - 2.3 Project Objectives
 - 2.4 Scope of the Project
3. Proposed System Overview
 - 3.1 System Users
 - 3.2 Major System Features
 - 3.3 System Workflow
4. Methodology
 - 4.1 Development Approach
 - 4.2 Requirement Analysis
 - 4.3 System Design
 - 4.4 Implementation
 - 4.5 Testing
 - 4.6 Evaluation
5. Technology and Development Environment
6. Database Design
7. Project Progress and Current Status
8. Team Member Responsibilities
9. Project Planning and Gantt Chart
10. Challenges and Solutions
11. Screenshots and System Demonstration
12. Future Work
13. Expected Outcome
14. Conclusion
15. References
16. Appendix

1. EXECUTIVE SUMMARY

FoodForward is a web-based food donation platform designed to connect donors, restaurants, and charitable organizations through a centralized digital system. The primary purpose of the project is to help reduce food waste and support people in need by making food donation easier, more transparent, and more efficient.

The platform allows donors to purchase meal packages from registered restaurants and donate them to verified charitable organizations. Donors can either select a specific charity or use the proposed **“Donate Where It's Needed Most”** feature, which allows the system to automatically assign donations to organizations with the highest current demand.

Restaurants are responsible for preparing the donated meal packages, while charitable organizations receive and confirm the donations. The system provides donation tracking, notification, donation history, and administrative monitoring facilities.

The project is currently in the development stage. The basic project structure, system requirements, technology stack, database planning, user roles, and major functional modules have been identified. Backend, frontend, database, and system integration activities are currently being developed and refined.

The overall project is estimated to be approximately **60% complete**. The remaining work focuses on completing the core features, integrating all system modules, testing the complete workflow, fixing identified issues, and preparing the system for final demonstration and deployment.

2. INTRODUCTION

2.1 Project Background

Food waste and food insecurity are two significant social problems. While large amounts of food are wasted every day, many individuals and communities still face difficulties accessing sufficient food.

Traditional food donation processes may involve multiple organizations and manual communication, making it difficult to track donations and determine where resources are most urgently required.

FoodForward proposes a digital solution to improve this process. The system connects donors, restaurants, and charitable organizations through a single web-based platform.

2.2 Problem Statement

- Lack of centralized donation management.
- Difficulty connecting donors with charities.
- Limited visibility of donation status.
- Manual communication between organizations.
- Difficulty identifying charities with the highest demand.
- Limited donation history and tracking.
- Lack of centralized administrative monitoring.
- Potential delays in delivering donated meals.

FoodForward aims to address these issues by developing an integrated web application that manages the complete donation process.

2.3 Project Objectives

- Develop a centralized food donation management platform.

- Connect donors, restaurants, and charities.
- Simplify the food donation process.
- Improve transparency in donation management.
- Provide donation tracking and status updates.
- Maintain donation history.
- Provide notifications for important donation activities.
- Allow charities to manage donation requests and confirmations.
- Provide administrators with system monitoring facilities.
- Implement automated donation allocation based on charity demand.

2.4 Scope of the Project

Donor Module: Registration, login, meal package browsing, donations, charity selection, automatic allocation, status tracking, and donation history.

Restaurant Module: Registration, profile management, meal package management, incoming donation orders, preparation status, and fulfillment.

Charity Module: Registration, profile management, incoming donations, confirmation, demand information, and donation requirements.

Administrator Module: User, restaurant, charity, organization verification, donation monitoring, status monitoring, system activity, and platform management.

3. PROPOSED SYSTEM OVERVIEW

3.1 System Users

Donor: Selects meal packages and makes donations to charitable organizations.

Restaurant: Prepares meal packages purchased through the platform for donation purposes.

Charity Organization: Receives donated meals and confirms successful receipt.

Administrator: Monitors and manages the complete platform.

3.2 Major System Features

- User registration and authentication.
- Role-based access control.
- Restaurant management.
- Charity management.
- Meal package management.
- Donation management.
- Automatic charity allocation.
- Donation tracking.
- Donation history.
- Notifications.

- Charity demand management.
- Administrative dashboard.

3.3 System Workflow

Donor Registration → Select Restaurant and Meal Package → Create Donation → Select Charity OR “Donate Where It's Needed Most” → Restaurant Receives Donation Order → Restaurant Prepares Meal Package → Meal Delivered to Charity → Charity Confirms Donation → Donation Status Updated → Donor Views Donation History

4. METHODOLOGY

4.1 Development Approach

The FoodForward project follows an iterative and incremental software development approach. The system is developed in separate modules, allowing the team to design, implement, test, and improve individual components before integrating the complete application.

The development process consists of: Requirement Analysis, System Planning, Database Design, System Design, Backend Development, Frontend Development, Module Integration, Testing, Evaluation, and Deployment Preparation.

4.2 Requirement Analysis

Functional Requirements: User registration and login; different user roles; restaurant meal packages; donor donations; charity selection; automatic allocation; donation confirmation; status tracking; donation history; notifications; and administrator monitoring.

Non-Functional Requirements: User-friendly, secure, responsive, reliable, scalable, maintainable, and efficient.

4.3 System Design

The system design focuses on defining architecture, database, user interfaces, workflows, and interactions between components. The major components are Authentication, Donor Management, Restaurant Management, Charity Management, Donation Management, Demand Management, Notification System, and Admin Management.

The system follows a three-layer concept: **Presentation Layer** (HTML, CSS, JavaScript, Bootstrap), **Application Layer** (Python and Django), and **Data Layer** (MySQL Database).

4.4 Implementation

The backend is being implemented using Python and Django. The frontend uses HTML, CSS, JavaScript, and Bootstrap. MySQL is used for database management. Git and GitHub are used for source-code version control and collaboration.

4.5 Testing

- Unit Testing: Individual components and functions.
- Integration Testing: Interaction between modules.
- System Testing: Complete integrated application.
- User Acceptance Testing: Evaluation from intended users.
- Security Testing: Authentication and access control.

4.6 Evaluation

- Functional correctness.
- Ease of use.
- System responsiveness.
- Data accuracy.
- Security.
- Donation tracking accuracy.
- Notification reliability.
- Effectiveness of automatic charity allocation.

5. TECHNOLOGY AND DEVELOPMENT ENVIRONMENT

Technology	Purpose
Python	Backend programming
Django	Web application framework
MySQL	Database management
HTML	Web page structure
CSS	Interface styling
JavaScript	Client-side functionality
Bootstrap	Responsive UI development
Git	Version control
GitHub	Source code repository

6. DATABASE DESIGN

The FoodForward database is designed using MySQL. The proposed major entities include User, Donor, Restaurant, Charity, Meal Package, Donation, Donation Status, Delivery, Notification, Charity Demand, and Donation History.

The database relationships are designed to allow the system to track a donation from the initial donor request through restaurant preparation and charity confirmation. The database also supports automatic allocation by maintaining information about the current demand levels of participating charities.

7. PROJECT PROGRESS AND CURRENT STATUS

Module	Status	Progress
Project Planning	Completed	100%
Requirement Analysis	Completed	100%
System Architecture	Completed	100%
Database Design	In Progress	80%
Django Backend	In Progress	65%
Frontend Development	In Progress	60%
Donor Module	In Progress	70%

Restaurant Module	In Progress	60%
Charity Module	In Progress	60%
Admin Module	In Progress	55%
Donation Tracking	In Progress	60%
Automatic Allocation	In Progress	40%
Notification System	Planned/In Progress	35%
Testing	Planned	20%
Deployment	Planned	0%

Overall Estimated Progress: 60%

8. TEAM MEMBER RESPONSIBILITIES

Team Member	Primary Responsibility	Current Tasks
[Member 1]	Project Coordinator / Backend Developer	Django backend, authentication, core business logic
[Member 2]	Frontend Developer / UI Designer	HTML, CSS, Bootstrap, JavaScript, responsive interface
[Member 3]	Database / Testing / Documentation	MySQL database, testing, documentation, project report

Project Coordinator: Coordinates activities, manages development progress, and integrates major modules.

Backend Developer: Develops Django models, views, authentication, APIs, and core logic.

Frontend Developer: Develops user interfaces and responsive design.

Database Developer: Designs and manages MySQL database structures and relationships.

Testing and Documentation: Performs testing, records issues, prepares documentation, and supports final evaluation.

9. PROJECT PLANNING AND GANTT CHART

The following proposed schedule can be adjusted according to the university semester timeline and actual project progress.

Activity	M1	M2	M3	M4	M5
Project Planning	■				
Requirement Analysis	■				
Literature Review	■	■			
System Design		■	■		
Database Development		■	■		
Backend Development		■	■	■	
Frontend Development		■	■	■	
Module Integration			■	■	
Donation Allocation			■	■	
Notification System			■	■	
System Testing				■	■
Bug Fixing				■	■
Final Evaluation					■

Documentation				■	■
Final Presentation					■

Phase 1: Planning and Requirements. **Phase 2:** System and Database Design. **Phase 3:** Backend and Frontend Development. **Phase 4:** Integration and Testing. **Phase 5:** Documentation, Evaluation, and Presentation.

10. CHALLENGES AND SOLUTIONS

Challenge 1 – Database Design: Designing relationships between donors, restaurants, charities, and donations is complex. **Solution:** A structured relational database model is used.

Challenge 2 – Multiple User Roles: Different users require different permissions. **Solution:** Role-based access control is being implemented.

Challenge 3 – Automatic Charity Allocation: Determining which charity needs donations most requires accurate demand information. **Solution:** Charity demand data will be maintained and used by predefined allocation criteria.

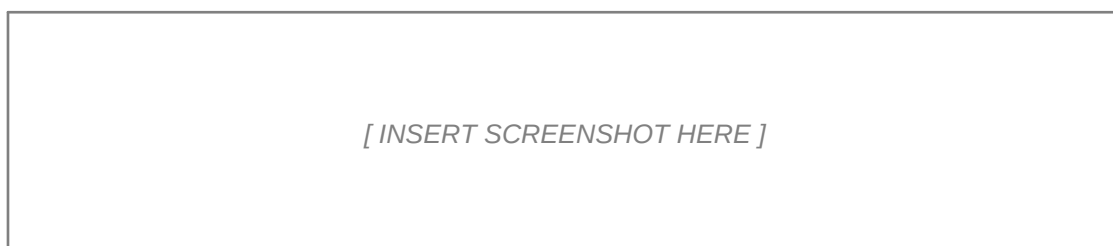
Challenge 4 – Donation Tracking: Tracking a donation through multiple stages is difficult. **Solution:** Donation status updates will track each donation from creation to confirmation.

Challenge 5 – System Integration: Integrating modules can cause compatibility issues. **Solution:** Modules are developed and tested incrementally before full integration.

11. SCREENSHOTS AND SYSTEM DEMONSTRATION

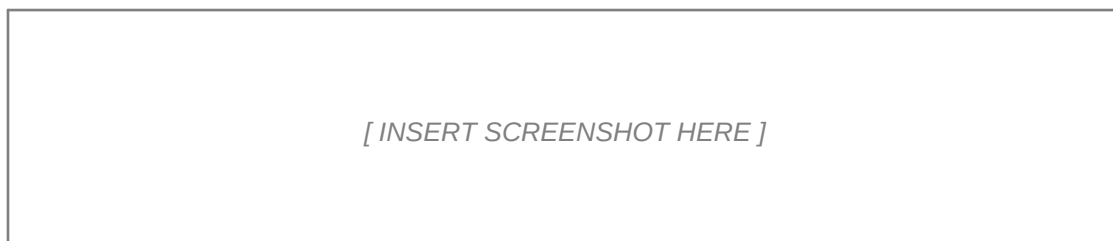
This section is reserved for screenshots of the actual FoodForward system. Insert the real screenshots from the current application in the placeholders below.

Figure 1: FoodForward Home Page



Homepage showing the main navigation and platform introduction.

Figure 2: User Registration Page



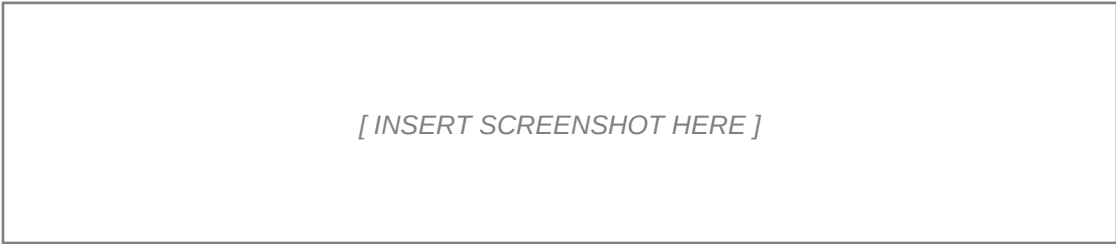
User registration interface for creating a FoodForward account.

Figure 3: Login Page



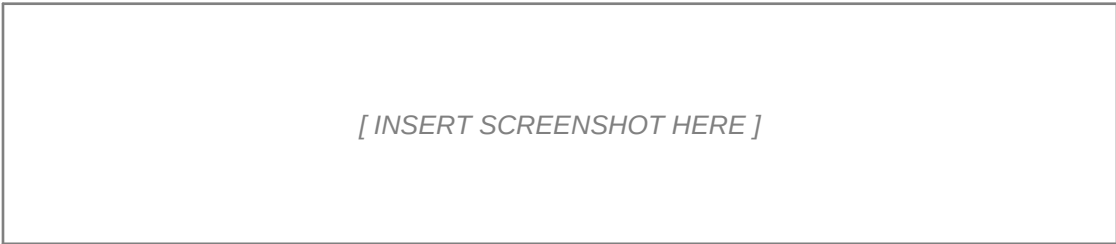
Secure login interface for accessing the platform.

Figure 4: Donor Dashboard



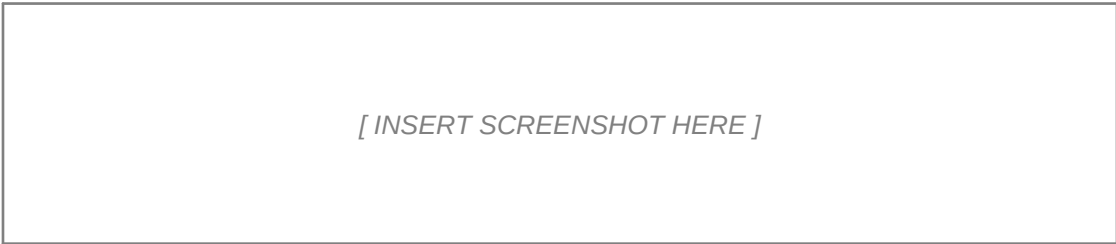
Donor dashboard showing available donation options and donation history.

Figure 5: Meal Package Selection



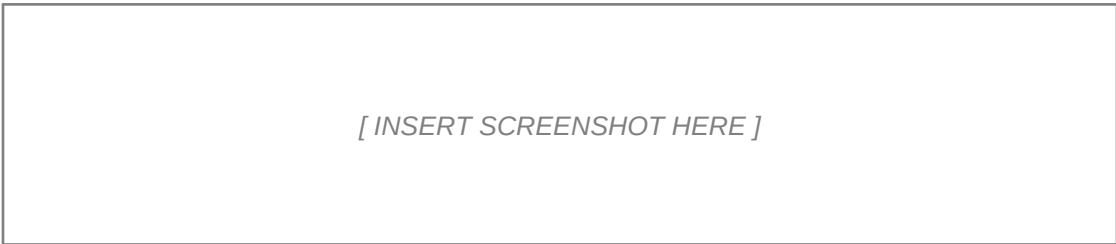
Interface for selecting meal packages from registered restaurants.

Figure 6: Charity Selection



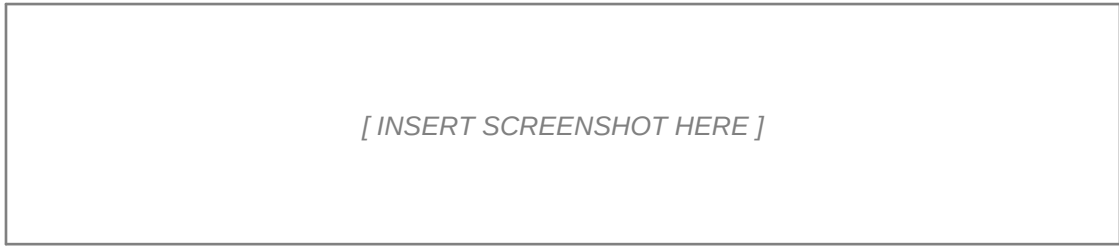
Interface allowing donors to select a charity for their donation.

Figure 7: Donate Where It's Needed Most



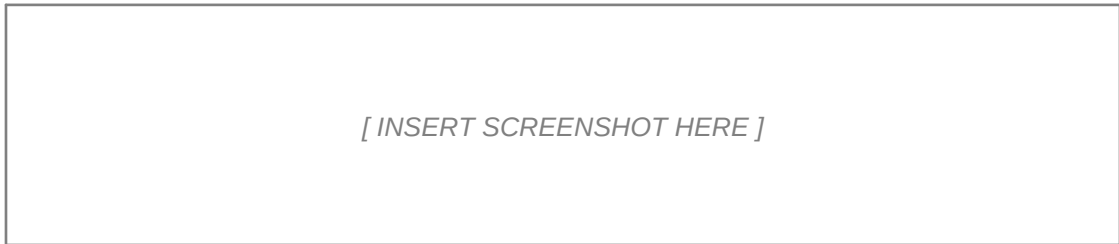
Automatic donation allocation feature based on current charity demand.

Figure 8: Restaurant Dashboard



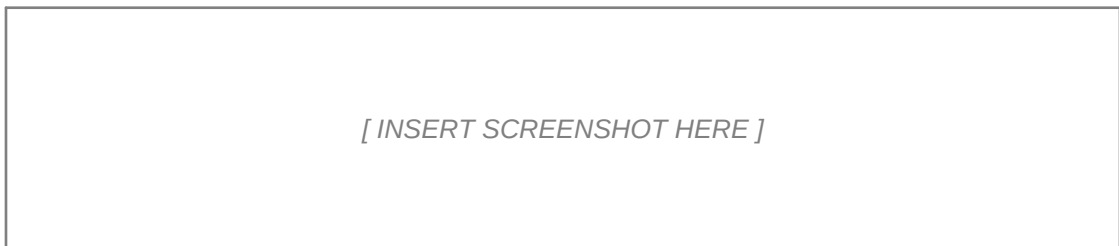
Restaurant dashboard for managing meal packages and donation orders.

Figure 9: Charity Dashboard



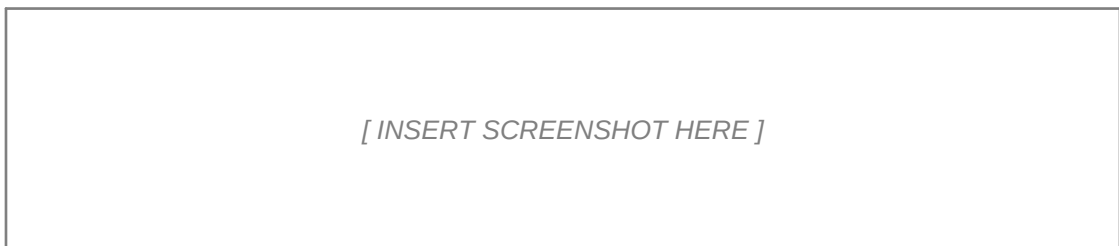
Charity dashboard for viewing and confirming incoming donations.

Figure 10: Admin Dashboard



Administrator dashboard for monitoring the FoodForward platform.

Figure 11: Donation Tracking



Donation tracking interface showing the current status of a donation.

12. FUTURE WORK

- Complete all database models.
- Complete backend development.
- Finalize donor, restaurant, and charity modules.
- Complete automatic charity allocation algorithm.
- Implement notification system.
- Complete administrator dashboard.
- Integrate all system modules.
- Conduct comprehensive system testing.

- Fix bugs and improve performance.
- Deploy the application.
- Conduct user acceptance testing.
- Complete final documentation.
- Prepare final project presentation.

13. EXPECTED OUTCOME

- Easier food donation.
- Better communication between stakeholders.
- Improved donation transparency.
- Efficient donation tracking.
- Better charity resource allocation.
- Improved administrative monitoring.
- Reduced food waste.
- Increased support for communities in need.

14. CONCLUSION

FoodForward is a web-based food donation platform developed to address challenges associated with food waste and food insecurity. The system connects donors, restaurants, and charitable organizations through a centralized digital platform.

The project has completed the planning and requirement analysis stages and has progressed into active development. The database, backend, frontend, and major functional modules are currently being developed.

The proposed “**Donate Where It's Needed Most**” feature is one of the key innovations of the platform. It is designed to improve donation distribution by directing resources toward organizations with higher current demand.

With completion of the remaining development, integration, and testing activities, FoodForward is expected to provide a practical and user-friendly solution for managing food donations.

15. REFERENCES

1. Django Software Foundation. (n.d.). *Django documentation*.
2. Python Software Foundation. (n.d.). *Python documentation*.
3. Oracle. (n.d.). *MySQL documentation*.
4. Bootstrap. (n.d.). *Bootstrap documentation*.
5. GitHub. (n.d.). *GitHub documentation*.
6. University project report and software engineering methodology guidelines consulted during report preparation.

16. APPENDIX

Appendix A: Team Contribution

Team Member	Planning	Design	Development	Testing	Documentation
[Member 1]	✓	✓	✓	✓	✓
[Member 2]	✓	✓	✓	✓	✓
[Member 3]	✓	✓	✓	✓	✓

Appendix B: Project Repository

GitHub Repository: [Insert GitHub Repository Link]

Appendix C: Additional Screenshots

- Database tables.
- Django admin panel.
- User authentication.
- Donation confirmation.
- Notification interface.
- Charity demand management.
- System testing results.

Appendix D: Supervisor Meeting Records

Meeting	Date	Discussion	Decision / Action
1	[Date]	Project proposal discussion	Project scope approved
2	[Date]	Requirement analysis	Core modules identified
3	[Date]	System design	Architecture reviewed
4	[Date]	Development progress	Backend and database reviewed
5	[Date]	Current progress	Remaining tasks identified